

MAT-113-1.00 - Introduction to Mathematical Thinking

Course Name		MAT-113-1.00	Course Title	Introduction to Mathematical Thinking
PS Course ID		669365	Base Program	BSMTH-2.00
Course Version		1.00	Credits	5
Academic Year Version First Offered		2025	School	School of Arts and Sciences
Academic Year Version Last Offered			Primary Course Author	Tulsi Srinivasan
Course Status		Active	Course Review Status	Approved
Availability Outside Base Program		Yes	Base Program for Requested Course	BSMTH-2.00
Requested Course Name		Introduction to Mathematical Thinking	Total Assessment Percentage	100
Curricular Group Information				
Curricular Group 1		Core	Curricular Group 2	
Curricular Group 3		Mathematics	Curricular Group 4	
Curricular Group 5				
Course Type Information				
Course Type 1		In person on campus	Course Type 2	English
Course Type 3		Foundational	Course Type 4	
Course Type 5				
Pre-requisite and Eligibility				
Prerequisites Course 1			Prerequisites Course 2	
Prerequisites Course 3			Prerequisites Course 4	
Prerequisites Course 5			Eligibility Criteria	
Any Other Exclusion				
Other Details				
Course delivery mode		Offline	Level	1
Number Of Teaching Hours		60		
UDL Details				
Alignment to UDL	No			
Description of UDL Alignment				
UDL Alignment	Click here to view the uploaded document !!			
Development Team				
Author				
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Abstract Description

This introductory course lays the foundation for all the mathematics courses that follow. It is taught using Inquiry-Based Learning (IBL): there are no lectures, there is no “covering” material in a textbook; instead students work together solving problems from carefully structured worksheets, and discuss, discover and raise questions. In school students do not get an opportunity to ‘do mathematics’. Further, there is a sharp discontinuity between school mathematics and university mathematics in content. While the former emphasises facility with concrete mathematical structures, the latter requires building intuition on abstract spaces, and working with entirely formally defined notions. This course attempts to fill this lacuna by both allowing students to experience the process of doing and creating mathematics. It is structured so that students explore new ideas and themes, make guesses, come up with new structures and terms, and then prove their results.

Description

Rationale and Introduction

Introduction to Mathematical Thinking is a course taken by mathematics majors in their first semester, and is part of a two-course introduction to mathematical thinking. This sequence of courses lays the foundation for all the mathematics courses that follow.

In this course, we introduce students to the act of "doing mathematics" and learning as a consequence of doing. But what does “doing mathematics” mean? It involves a range of processes such as abstraction, representation, reasoning, argumentation, visualisation, estimation, approximation, pattern recognition, symbolisation, making connections, formalisation, and so on. It also involves activities such as choosing between multiple representations, examining extreme cases and typical ones to come up with conjectures, actively searching for counterexamples, looking for invariances, generalising to simplify problems and building on answers to generate more questions.

Further, there is a sharp discontinuity between school mathematics and university mathematics in content. While the former emphasises facility with mathematical structures seen in the world, the latter requires building intuition on abstract spaces, and working entirely with formally defined notions. This discontinuity is sharp in the practice of reading and writing mathematics, as university mathematics is entirely discussed in the disciplinary language of mathematics, that is stylised and formal.

This course attempts to fill this lacuna by both allowing students to experience the process of doing and creating mathematics, using Inquiry-Based Learning (IBL). It is structured so that students explore new ideas and themes, make guesses, come up with new structures/terms, and then prove their results. The content is chosen so as to be foundational, within reach of all students, and to showcase a variety of problem solving techniques and mathematical processes from different areas of maths.

Intended Learning Outcomes	
Intended Learning Outcome	Description
ILO 1	State and explain key definitions and results arising from the exploratory exercises from the course
ILO 2	Pose questions and formulate conjectures in mathematical terms based on observations of patterns
ILO 3	Comprehend and explain mathematical arguments related to the exploratory exercises of the course.
ILO 4	Write mathematical arguments using appropriate mathematical language.
ILO 5	Appreciate the necessity for abstraction, precise language and formal proofs in mathematics.
ILO 6	Listen to and process arguments given by peers, and respond to them in a manner conducive to collaboration.

Pedagogy	
Pedagogical Practice	Pedagogy Description
Inquiry based learning	This course is taught principally through Inquiry-Based Learning (IBL). Students are encouraged to come up with observations and questions through motivating problems. They then work through a problem set in groups without consulting any external material, mostly outside class, and each group shares its solutions with the class. The class discusses the solution and moves on to the next problem only when everyone is satisfied with the solution. Following this, students return to the motivating problems and present their proofs in the same format. This process of “doing mathematics” is graded. Group-work is chosen over individual work in this course so students get comfortable working together, and to provide an intermediate stage in which they can present their ideas before sharing them with the whole class. Each group will be asked to spend at least two hours outside the classroom continuing work on the problem sheets. Students are not allowed to read about topics from other sources after they finish each unit, but are strongly encouraged to do so once each unit is completed. Except for specific topics that are covered through lectures, the instructor acts as a facilitator, only intervening when necessary, and summarising or reviewing important results as they are proved. Since the implementation of IBL depends on the class and instructor, there will be variation from one section and one offering of the course to another. The instructor may use their discretion to let the class spend more time on a topic that they find hard and convert some other topic to a lecture, or simply drop it. The value of the course lies less in the amount of content covered (as most of the concepts here will be revisited in future courses), than in the process of learning and discovery.

Other	The ratio of classes to tutorials is 3:1. Class time is spent mostly on student presentations. The tutorial hours will be spent on reviews and supplementary exercises, and groups can sometimes use the time to work on problems together.
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Assessment			
Total Assessment Percentage			100
Assessment Type	Number of ILO Linkage	Assessment Percentage	Description
In-class exam	3	20	Cumulative final test
In-class exam	3	20	Midterm test
Take home assignment	3	30	Students will submit regular assignments based on the problem sheets and the reading and writing exercises. These will be graded for mathematical completeness and correctness, as well as for clarity of writing. Regular low-stakes assessments allow for frequent feedback from the instructor, and help the student plan their studies outside class. Assignments will typically consist of 2-3 short questions.
Group presentation	4	30	Each student will be assigned a weekly presentation/discussion grade. Presentation grades are typically the same for all group members, independent of who makes the presentation. Discussion grades are individual. Presentations with incomplete ideas and genuine efforts in a wrong direction are given a higher grade than making no attempt.

Syllabus Summary

The course is taught through IBL. Students will be asked to work their way through a series of problems, through which they are guided through the course material. The problems consist of exploratory exercises, looking at examples and finding patterns, and then proving observations.

A brief overview of the units is given below:

Unit no.	Unit name	Topics	Weeks	Assessments
1	Introduction	Knots and tangles, the process of mathematics	1-2	Assignment 1
2	Elementary number theory	Greatest common divisor, Euclidean algorithm, primes, Bezout's lemma	3-7	Assignments 2-5
			8	Midterm test
3	Continued fractions	Definitions, applications in approximations and solutions to some diophantine equations	9-11	Assignments 6-7
4	Pythagorean triples	Explorations around Pythagorean triples	12	Assignment 8
5	Rational tangles		13-14	Assignment 9-10
			15	Final test

The choice of pedagogy is to allow students to experience what it means to discover mathematical truths, so they are asked not to refer to text books and other sources until the main results are proven at the end of the third unit. The readings given during the course will be created by the course developers.

Each unit begins with a few challenging problems to motivate the ideas and techniques explored in the unit. The worksheets will guide the students through exploratory problems, towards important mathematical techniques, processes and results, which can be used to approach the initial questions.

A tentative week-by-week schedule is given in this course document. However, the choice of pedagogy means there will be variation from one section and one offering of the course to another. The instructor may use their discretion to let the class spend more time on a topic that they find hard and convert some other topic to a lecture, or simply drop it. The value of the course lies less in the amount of content covered than in the process of learning and discovery.

Syllabus

Unit Name

Unit 1: Introduction

Description

This short unit introduces students to the process of doing mathematics collaboratively. Knots and tangles used to illustrate the process of coming up with hypotheses, mathematising ideas and trying to justify them.

Week 1: Introduction to the course and to knots and tangles.

Week 2: Some simple invariants for telling knots and links apart: *linking number* of links, and *tricolorability* of knots. Time permitting, we'll take a look at the *unknotting number*.

Readings

References and recommended readings:

Colin Adams, *The Knot Book*, Chapter 1, Time Permitting: Portions of Chapters 2 and Chapter 3

Unit Name

Unit 2: Elementary number theory

Description

This unit covers the building blocks for the course – the division algorithm, Euclid's algorithm and unique prime factorisation. Mathematical induction and its equivalent statements are done through a lecture.

Week 3: Exploration of the Euclidean algorithm through paper-folding, divisibility, gcd.

Week 4-5: Proofs of the division and Euclidean algorithm. Mathematical induction is covered in a lecture.

Week 6-7: Prime numbers, coprime numbers, proof of unique prime factorisation

Week 8: Review and midterm test

Readings

References and recommended readings:

1. David Burton, *Elementary Number Theory*, McGraw Hill, 2011, Chapter 2 and Section 3.1, pp 13-44
2. Kenneth Rosen, *Elementary Number Theory and its Applications*, Addison-Wesley, 1986. Chapters 1-3.
3. Wacław Sierpinski, *250 Problems in Elementary Number Theory*, Elsevier 1970. Chapters 1, 2 and 4.

Unit Name

Unit 3: Continued fractions

Description

Building on Unit 2, this unit looks at the representation of numbers as continued fractions, and some applications of this representation.

Week 9: Return to the paper folding-algorithm for gcd; finite continued fractions and their connection with rational numbers, convergents of a fraction.

Week 10: Applications of continued fractions in solving linear diophantine equations. Infinite continued fractions, approximations of irrationals

Week 11: Further applications: Pell's equation, calendar problems, etc.

Readings

References and recommended readings:

1. David Burton, *Elementary Number Theory*, McGraw Hill, 2011, Chapter 15, pp 303-352
2. Allan Hatcher. *Topology of Numbers*, AMS, 2024, Sections 2.1 and 2.2, pp 36-55

Unit Name

Unit 4: Pythagorean triples

Description

The question of finding Pythagorean triples is introduced as another example of a diophantine equation. Students are invited to pose and investigate their own questions..

Week 12: Introduction to Pythagorean triples. Open-ended investigation of similar questions.

Readings

References and recommended readings:

1. David Burton, *Elementary Number Theory*, McGraw Hill, 2011, Chapter 12, pp 245-250
2. Allan Hatcher. *Topology of Numbers*, AMS, 2024, Chapter 0, pp 1-26

Unit Name

Unit 5: Rational tangles

Description

This unit introduces students to rational tangles through a hands-on activity. Students observe the correspondence between rational numbers and rational tangles, and then use continued fractions to prove the correspondence holds. This unit is a follow up to the continued fractions unit, and gives a nice climax to the course.

Week 13: Introduction to rational tangles through hands-on exploration using a ‘tangle board’ and string, correspondence between rational numbers and rational tangles.

Week 14: Using the continued fraction decomposition of a rational number to construct a tangle, proof of the one-to-one correspondence between rationals and rational tangles

Readings

References and recommended readings:

1. James Tanton, *Understanding Rational Tangles*, <https://reuf.aimath.org/assets/session-materials/JTantonRationalTangles.pdf>
2. Tom Davis, *Conway’s Rational Tangles*, <http://www.geometer.org/mathcircles/tangle.pdf>

Annexure

ILO Assessment Linkage

ILO	Assessment	Assessment Notes
ILO 1	In-class exam	Cumulative final test
	In-class exam	Midterm test
ILO 2	Group presentation	Each student will be assigned a weekly presentation/discussion grade. Presentation grades are typically the same for all group members, independent of who makes the presentation. Discussion grades are individual. Presentations with incomplete ideas and genuine efforts in a wrong direction are given a higher grade than making no attempt.
	In-class exam	Midterm test
	Take home assignment	Students will submit regular assignments based on the problem sheets and the reading and writing exercises. These will be graded for mathematical completeness and correctness, as well as for clarity of writing. Regular low-stakes assessments allow for frequent feedback from the instructor, and help the student plan their studies outside class. Assignments will typically consist of 2-3 short questions.
	In-class exam	Cumulative final test
ILO 3	Group presentation	Each student will be assigned a weekly presentation/discussion grade. Presentation grades are typically the same for all group members, independent of who makes the presentation. Discussion grades are individual. Presentations with incomplete ideas and genuine efforts in a wrong direction are given a higher grade than making no attempt.
	Take home assignment	Students will submit regular assignments based on the problem sheets and the reading and writing exercises. These will be graded for mathematical completeness and correctness, as well as for clarity of writing. Regular low-stakes assessments allow for frequent feedback from the instructor, and help the student plan their studies outside class. Assignments will typically consist of 2-3 short questions.
		Cumulative final test
	In-class exam	Midterm test

ILO 4	In-class exam Take home assignment	Students will submit regular assignments based on the problem sheets and the reading and writing exercises. These will be graded for mathematical completeness and correctness, as well as for clarity of writing. Regular low-stakes assessments allow for frequent feedback from the instructor, and help the student plan their studies outside class. Assignments will typically consist of 2-3 short questions.
ILO 5	Group presentation	Each student will be assigned a weekly presentation/discussion grade. Presentation grades are typically the same for all group members, independent of who makes the presentation. Discussion grades are individual. Presentations with incomplete ideas and genuine efforts in a wrong direction are given a higher grade than making no attempt.
ILO 6	Group presentation	Each student will be assigned a weekly presentation/discussion grade. Presentation grades are typically the same for all group members, independent of who makes the presentation. Discussion grades are individual. Presentations with incomplete ideas and genuine efforts in a wrong direction are given a higher grade than making no attempt.

Assessment ILO Linkage

Assessment	Assessment Notes	ILO
Group presentation	Each student will be assigned a weekly presentation/discussion grade. Presentation grades are typically the same for all group members, independent of who makes the presentation. Discussion grades are individual. Presentations with incomplete ideas and genuine efforts in a wrong direction are given a higher grade than making no attempt.	ILO 2 ILO 3 ILO 5 ILO 6
In-class exam	Cumulative final test	ILO 1 ILO 4 ILO 2
In-class exam	Midterm test	ILO 2 ILO 1 ILO 4
Take home assignment	Students will submit regular assignments based on the problem sheets and the reading and writing exercises. These will be graded for mathematical completeness and correctness, as well as for clarity of writing. Regular low-stakes assessments allow for frequent feedback from the instructor, and help the student plan their studies outside class. Assignments will typically consist of 2-3 short questions.	ILO 2 ILO 3 ILO 4